

Ch 6: Control and Coordination

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. What is control and coordination? Why is it necessary in multicellular organisms? (PYQ 2019)

Q2. What is stimulus? Give two examples of stimuli and the response produced by organisms. (PYQ 2020)

Q3. Define reflex action. Give one example of a reflex action in humans. (PYQ 2022)

Q4. What is a reflex arc? Name the main parts involved in a reflex arc. (PYQ 2019)

Q5. What is a neuron? Name its three main parts and state the function of each part. (PYQ 2022)

Q6. What is a synapse? How does an impulse pass from one neuron to another across a synapse? (PYQ 2020)

Q7. Name the two main parts of the human nervous system. What is the role of the central nervous system? (PYQ 2019)

Q8. What is the function of the brain? Name the three main parts of the brain. (PYQ 2022)

Q9. What is the function of the cerebrum? Which part of the brain controls thinking and memory? (PYQ 2020)

Q10. What is the role of the cerebellum? Name any two activities controlled by the cerebellum. (PYQ 2019)

Q11. What is the function of the medulla? Name two involuntary actions controlled by it. (PYQ 2022)

Q12. What are voluntary actions and involuntary actions? Give one example of each. (PYQ 2020)

Q13. Why are reflex actions faster than normal voluntary actions? (PYQ 2019)

Q14. What is the spinal cord? State two functions of the spinal cord. (PYQ 2022)

Q15. What are receptors? Name any two sense organs and the receptors present in them. (PYQ 2020)

Q16. What are hormones? Name the type of glands that secrete hormones. (PYQ 2019)

Q17. What is the endocrine system? Name any two endocrine glands in humans. (PYQ 2022)

Q18. Name the hormone secreted by the pituitary gland. Why is pituitary called the master gland? (PYQ 2020)

Q19. Name the hormone secreted by the thyroid gland. What is the function of thyroxine? (PYQ 2019)

Q20. Why is iodine necessary in our diet? Name the disease caused due to iodine deficiency. (PYQ 2022)

Q21. Name the hormone that regulates blood sugar level. Which gland secretes it? (PYQ 2020)

Q22. What is adrenaline? Which gland secretes it and when is it secreted? (PYQ 2019)

Q23. State two effects of adrenaline on the human body during emergency situations. (PYQ 2022)

Q24. Name the male and female sex hormones. State one function of each. (PYQ 2020)

Q25. What is auxin? How does auxin help in the growth of plant shoots? (PYQ 2019)

- Q26. What is phototropism? Give one example of positive phototropism. (PYQ 2022)**
- Q27. What is geotropism? Give one example each of positive and negative geotropism. (PYQ 2020)**
- Q28. What is hydrotropism? Which part of the plant shows hydrotropism? (PYQ 2019)**
- Q29. What is chemotropism? Give one example from plant reproduction. (PYQ 2022)**
- Q30. What is nastic movement? How is it different from tropic movement? (PYQ 2020)**
- Q31. Why do leaves of Mimosa pudica fold when touched? Name this movement. (PYQ 2024)**
- Q32. Name the plant hormones responsible for: (i) cell elongation (ii) cell division (iii) growth inhibition (iv) wilting of leaves. (PYQ 2019)**

SECTION B - 3 MARKS

- Q1. Explain the need for control and coordination in living organisms. Why do animals require a faster system of coordination than plants? (PYQ 2019)**
- Q2. Draw a labelled diagram of a neuron and explain the function of dendrite, cell body, axon, and nerve ending. (PYQ 2020)**
- Q3. Explain how a nerve impulse travels through a neuron. What is the role of chemical signals at the synapse? (PYQ 2022)**
- Q4. Describe the reflex arc with an example. Why is the spinal cord involved in reflex actions? (PYQ 2019)**
- Q5. Compare voluntary and involuntary actions: (PYQ 2022)**

Basis of Comparison	Voluntary Actions	Involuntary Actions
Control		
Speed		
Conscious effort		
Examples		

Q6. Explain the structure and function of the human brain. Mention the role of cerebrum, cerebellum, and medulla. (PYQ 2020)

Q7. What are receptors? Explain the role of receptors in detecting changes in the environment with suitable examples. (PYQ 2019)

Q8. Compare nervous coordination and hormonal coordination: (PYQ 2022)

Basis **Nervous Coordination** **Hormonal Coordination**

Nature of signal

Speed of response

Duration of effect

Pathway

Example

Q9. What are endocrine glands? How are endocrine glands different from exocrine glands? Give two examples of each. (PYQ 2020)

Q10. Explain the role of the pituitary gland, thyroid gland, and adrenal gland in human beings. (PYQ 2019)

Q11. Explain how insulin controls blood sugar level. What happens when the pancreas secretes insufficient insulin? (PYQ 2022)

Q12. What is adrenaline called the emergency hormone? Explain the changes produced in the body when adrenaline is secreted. (PYQ 2020)

Q13. Explain feedback mechanism in hormone regulation with the example of thyroxine secretion. (PYQ 2019)

Q14. What are plant hormones? State the functions of auxin, gibberellin, cytokinin, and abscisic acid. (PYQ 2022)

Q15. Explain phototropism in plants with the help of auxin distribution. Why does the shoot bend towards light? (PYQ 2020)

Q16. Describe an activity to show phototropism in plants. What conclusion is obtained from the activity? (PYQ 2019)

Q17. Compare tropic and nastic movements: (PYQ 2022)

Basis **Tropic Movement** **Nastic Movement**

Direction of stimulus

Direction of response

Growth involved?

Speed

Example

Q18. What are tropic movements? Explain phototropism, geotropism, hydrotropism, and chemotropism with examples. (PYQ 2020)

Q19. Explain the movement of tendrils around a support. Which type of plant movement is shown here? (PYQ 2019)

Q20. Why is chemical coordination necessary in plants? How do plant hormones help in growth and development? (PYQ 2022)

Q21. Explain how the folding of leaves in *Mimosa pudica* differs from the bending of a plant shoot towards light. (PYQ 2020)

Q22. What happens when: (i) iodine is deficient in diet (ii) insulin is not secreted properly (iii) adrenaline is released in excess during fear? (PYQ 2019)

Q23. Write the functions of the following hormones: (PYQ 2022)

Hormone Gland / Source Function

Thyroxine

Insulin

Adrenaline

Testosterone

Oestrogen

Q24. Describe how coordination in plants is different from coordination in animals. Give three points of difference. (PYQ 2020)

Q25. Explain why a person withdraws the hand immediately after touching a hot object. Trace the pathway of the impulse. (PYQ 2019)

Q26. Why are hormones secreted in small quantities but produce large effects? Explain with an example. (PYQ 2022)

Q27. Explain the role of the hypothalamus and pituitary gland in coordinating endocrine activity. (PYQ 2020)

Q28. Describe the response of plant roots towards water and gravity. How are these responses useful to the plant? (PYQ 2019)

Q29. Write the differences between growth movement and non-growth movement in plants with one example each. (PYQ 2022)

Q30. What is meant by coordination between nervous system and endocrine system? Explain with the example of stress response. (PYQ 2020)

SECTION C - 5 MARKS

Q1. Describe the human nervous system in detail. Explain the central nervous system, peripheral nervous system, and the role of the brain and spinal cord in coordination. (PYQ 2022)

Q2. Draw a labelled diagram of a neuron. Explain the structure of a neuron and describe how information is transmitted from receptors to effectors. (PYQ 2020)

Q3. Explain reflex action with the help of a labelled reflex arc. Describe the pathway of impulse when a person touches a hot object. Why are reflex actions important? (PYQ 2019)

Q4. Describe the structure and functions of the human brain. Explain the functions of cerebrum, cerebellum, medulla, and hypothalamus. (PYQ 2022)

Q5. Compare nervous coordination and hormonal coordination in detail. Mention five differences and explain why both systems are needed in the human body. (PYQ 2020)

Q6. Describe the endocrine system in humans. Explain the location and function of pituitary, thyroid, pancreas, adrenal gland, testes, and ovaries. (PYQ 2019)

Q7. Explain feedback mechanism in hormonal control. Describe how thyroxine level is regulated in the body and why iodine is essential for this process. (PYQ 2022)

Q8. Explain the role of adrenaline during emergency situations. Describe the changes in heart rate, breathing rate, blood supply, and energy availability during fear or stress. (PYQ 2020)

Q9. Write a detailed comparison of the following human hormones: (PYQ 2019)

Hormone	Endocrine Gland	Main Function	Deficiency / Disorder
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Thyroxine Thyroid gland

Insulin Pancreas

Adrenaline Adrenal gland

Growth hormone Pituitary gland

Sex hormones Testes / Ovaries

Q10. Describe coordination in plants. Explain how plants respond to light, gravity, water, chemicals, and touch with suitable examples. (PYQ 2022)

Q11. Explain phototropism in detail. Describe the role of auxin in bending of shoots towards light and roots away from light. Draw a simple diagram to support your answer. (PYQ 2020)

Q12. Compare tropic movements and nastic movements in plants. Explain phototropism, geotropism, hydrotropism, chemotropism, thigmotropism, and the folding of Mimosa leaves. (PYQ 2019)

Q13. Write a comprehensive note on plant hormones covering auxins, gibberellins, cytokinins, abscisic acid, and ethylene. Mention their major functions in plant growth and development. (PYQ 2022)

Q14. Case Study: A student kept a potted plant near a window. After a few days, the shoot bent towards the window while the roots continued growing downwards. (PYQ 2024 case-based)

Question Answer

Which movement is shown by the shoot?

Which hormone is responsible for the bending of the shoot?

Why does the shoot bend towards light?

Which movement is shown by the root?

How are these movements useful to the plant?

Q15. Answer the following questions related to control and coordination: (PYQ 2023)

Question Answer

Why does a nerve impulse travel in one direction across a synapse?

Why is the pituitary gland called the master gland?

What would happen if insulin is not produced in sufficient amount?

Why are hormones called chemical messengers?

How is movement in Mimosa different from movement of a tendril?

Q16. Draw and explain any two diagrams from this chapter: (i) neuron (ii) reflex arc (iii) human brain (iv) endocrine glands in humans. Also write the function of each labelled part. (PYQ 2022 comprehensive)